

## Shilu Shrestha

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### PROFILE SUMMARY

Highly experienced Software Developer (12+ years) currently pursuing an MSc in Artificial Intelligence at the University of Bath. Proven track record in full-stack development and system architecture, now pivoting into AI-driven Business Transformation. Seeking to leverage extensive technical maturity and academic specialisation in AI to drive innovation.

### EDUCATION

#### **Masters of Science in Artificial Intelligence (University of Bath)**

Sept 2025 – Present | Bath, United Kingdom

Key Modules: Applied AI, Machine Learning, Neural Networks, Deep Learning, Reinforcement Learning, Probability & Statistics, Data Analysis

Focus: Gaining a solid understanding of deep learning fundamentals, architectures, optimisation and evaluation.

### PROJECTS

#### Multi-Agent Reinforcement Learning (MARL)

*Developed a cooperative multi-agents system to learn to optimise resource allocation in dynamic warehouse environment.*

Python, TensorFlow, Data Structures and Algorithms, Optimisation, Scalability

#### Convolutional Neural Networks For Age and Gender Classification

*Engineered and tuned Convolutional Neural Networks for image recognition tasks, achieving 87% accuracy with advance hyper-parameter optimisation.*

TensorFlow, Keras, Image Augmentation, Python, Performance Tuning

#### **Bachelors of Engineer in Information Technology (Pokhara University)**

Aug 2008 – Oct 2012 | Kathmandu, Nepal

CGPA: 3.27/4

### TECHNICAL SKILLS

**Machine Learning and AI:** Data Preprocessing, Classical ML (Scikit-learn), Deep Learning (TensorFlow, PyTorch), Model Training & Evaluation, Generative AI (OpenAI APIs, LangChain), Modal Optimisation and Scaling

**Programming & Development:** Python, JavaScript & TypeScript, React, React Native, SQL, NoSQL, REST APIs, GraphQL, Git, CI/CD, C++

**Database and Cloud:** PostgreSQL, MongoDB, Firebase, Supabase, Cloud Functions, Cloud Integration (AWS / GCP basics), ML Infrastructure basics

**Tools & Frameworks:** NumPy, Pandas, Matplotlib, FastAPI, Flask, Jupyter, VS Code, Cursor, Power BI

### WORK EXPERIENCE

Danphe Software Labs

Aug 2024 - Oct 2025

Senior Software Engineer

- Designed and implemented AI controls for proprietary **LLM Integration (GPT-4o) and embedding models** for intelligent search and content generation, leveraging data for strategic insights.
- Established testing frameworks and **prompt-engineering protocols for RAG pipelines** to enhance system performance and user experience.
- Optimised app performance by identifying bottlenecks, cutting load times by 25% to improve UX.
- Designed and implemented a robust multi-language support system to increase accessibility and expand the CRM's reach across global markets.
- Provided technical leadership for a team of 7-8 engineers, conducting code reviews and fostering a collaborative environment to meet tight deadlines and deliver high-quality solutions.

Career Break

May 2023 - Apr 2024

- Volunteered as a Section Leader for Stanford University's Code in Place, applying **Python scripting** expertise and demonstrating strong communication skills.
- Developed an MVP mobile application, for child chore and allowance management using React Native, TypeScript, data storage controls (Firebase) and Cloud Functions security.

Fleetpanda

San Francisco, USA (Remote)

Senior Software Engineer

Jun 2018 – Apr 2023

- Engineered and maintained robust software environment services for real-time data monitoring and control of Liquid Control Registers ensuring reduced driver's operational time by 32%.
- Designed an offline-first architecture using Apollo Client to ensure data continuity and system resilience in challenging environment mitigating operational risks from network failures (distributed system design).
- Led cross-functional Scrum teams (18+ engineers) through full-cycle delivery, facilitating Sprints, Daily Stand-ups, and Retrospectives to drive process standardisation and reduce time-to-market.
- Managed and automated CI/CD pipelines (Travis CI) and testing (Jest, TestFlight, Android Beta), ensuring change control and high release quality.

Chitooo, Leapfrog Technology, CedarGate, Blue Ocean Systems

Kathmandu, Nepal

Foundational Engineering Delivery and Leadership (2012-2018)

- Defined technical vision as Chief Technology Officer(CTO) and stakeholder management, leading end-to-end software development across cross-functional teams mentoring and ensuring adherence to engineering lifecycle and design principles (SOLID, DRY, KISS) to manage project complexity and quality.
- Developed high-performance applications, consistently demonstrating strong technical skills and attention to details in application development and data handling.

CERTIFICATIONS

|                                                  |         |
|--------------------------------------------------|---------|
| <a href="#">Advance React</a>                    | Scrimba |
| <i>Skills: Reusability, Routing, Performance</i> |         |
| <i>June 2024</i>                                 |         |
| <a href="#">Advance React</a>                    | Meta    |
| <i>Skills: Custom Hooks, Unit Testing, JSX</i>   |         |
| <i>June 2024</i>                                 |         |